# BST Vol 7 Ch 10 — EM in Matter: Dispersion, Optics, Refractive Indices (v0.4.1, substantive 3-level pedagogy upgrade)

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2026-05-23 Saturday

## Vol 7 Chapter 10 — EM in Matter

### Headline result

EM propagation in materials is governed by frequency-dependent dielectric constant $\varepsilon(\omega)$ and permeability $\mu(\omega)$:

$$n(\omega) = \sqrt{\varepsilon(\omega)\mu(\omega)}$$

with phase velocity v = c/n. Lorentz oscillator model captures dielectric response:

$$\varepsilon(\omega) = 1 + \sum_j \frac{\omega_{pj}^2}{\omega_j^2 - \omega^2 - i\gamma_j\omega}$$

BST identification: oscillator frequencies $\omega$_j inherit substrate K-type Casimir spectrum framework (Vol 1 Ch 5 + T2435). Refractive indices at BST primary ratios predicted for selected materials — substrate-natural photonic resonances testable at $10K photonic crystal falsifier (Vol 9 Ch 9 cheapest BST falsifier).

### Substrate mechanism

Lorentz oscillator model:

Treating atoms/molecules as damped harmonic oscillators driven by EM field:

$$m\ddot{x} + m\gamma\dot{x} + m\omega_0^2 x = -eE$$

Steady-state solution → frequency-dependent polarization → dielectric function $\varepsilon(\omega)$ with resonance at $\omega$_0.

K-type Casimir spectrum framework (T2435):

Per Lyra Saturday operator zoo: K-type Casimir = $C_2$ = 6 substrate ground state. Oscillator frequencies $\omega_j$ in dielectric response inherit substrate framework: - Resonances at substrate-natural energy scales - BST primary integer ratios in selected materials - Cross-link Vol 9 Ch 9: photonic crystals at BST primary lattice constants

Refractive index BST primary forms:

For substrate-natural lattice materials, $n(\omega)$ at specific BST primary frequency ratios should show enhanced response. $10K photonic crystal experimental program (Vol 9 Ch 9) is the cheapest BST falsifier.

## Optical phenomena

Reflection + transmission: Fresnel equations at boundary; reflection coefficient depends on $(n_1 - n_2)/(n_1 + n_2)$.

Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$ from boundary conditions.

Dispersion: $n(\omega)$ varies → prism + rainbow spectral separation.

Absorption + extinction: Im(n) gives absorption coefficient $\alpha_{abs} = 2\omega\, \mathrm{Im}(n)/c$.

Anomalous dispersion: near resonance, $n(\omega)$ decreases with $\omega$ (vs normal increasing); related to Kramers-Kronig dispersion relations.

## Match precision

I-tier framework on substrate-optics connection. Standard optics + dispersion theory preserved at any precision. Substrate-natural refractive index predictions at $10K photonic crystal falsifier (Vol 9 Ch 9).

## Cross-volume dependencies

- Vol 9 Ch 3 (Electron Band Structure) — dielectric response in solids
- Vol 9 Ch 9 (Photonic Crystals) — $10K cheapest substrate falsifier
- Vol 1 Ch 5 (Casimir Algebra) — K-type oscillator framework
- Vol 7 Ch 5 (EM Waves) — wave propagation in vacuum
- Vol 8 Ch 8 (Oscillations) — driven harmonic oscillator framework

## K-audit anchor

K226 Vol 7 Ch 10 EM in Matter K-audit pre-stage (Keeper pending).

## Pedagogical walkthrough (3-level)

### Level 1 — Bright 5th-grader

> When light enters glass or water, it slows down! The "refractive index" n tells how much: for glass n ≈ 1.5 (light goes 1.5× slower in glass than vacuum); for water n ≈ 1.33.
>
> Different colors of light slow down by different amounts — this is why prisms split white light into rainbow colors!
>
> BST predicts: certain "magic" refractive indices, set by BST primary integers (2, 3, 5, 7), should be substrate-natural — and you can test this with $10K of photonic crystals (Vol 9 Ch 9). This is the cheapest BST falsifier in physics!

### Level 2 — Undergraduate physics student

Dielectric function from Lorentz oscillator model:

Driven damped oscillator (electron in atom):

$$m\ddot{x} + m\gamma\dot{x} + m\omega_0^2 x = -eE_0 e^{-i\omega t}$$

Solution: x(t) = -eE_0/(m($\omega$_0$^2$ - $\omega^2$ - i$\gamma\omega$)) e^{-i$\omega$t}.

Polarization P = -nex (n electrons per volume) → dielectric function:

$$\varepsilon(\omega) = 1 + \frac{ne^2/(m\varepsilon_0)}{\omega_0^2 - \omega^2 - i\gamma\omega}$$

Refractive index: n($\omega$) = $\sqrt{}$($\varepsilon$($\omega$)$\mu$($\omega$)). For non-magnetic ($\mu$ = 1): n = $\sqrt{\varepsilon}$.

Real + imaginary parts: - Re($\varepsilon$): refraction (phase velocity c/n) - Im($\varepsilon$): absorption

Kramers-Kronig relations: dispersion relations connect Re($\varepsilon$) and Im($\varepsilon$) via Hilbert transform.

Examples: - Water visible: n ≈ 1.33 - Glass visible: n ≈ 1.5 - Diamond visible: n ≈ 2.4 - Silicon visible: n ≈ 3.5 (highly dispersive near band-edge)

BST framework: - Oscillator frequencies $\omega$_j inherit substrate K-type Casimir spectrum - BST primary ratios at substrate-natural materials - Photonic crystal $10K falsifier (Vol 9 Ch 9)

### Level 3 — Graduate physics student / theorem-level

Lorentz oscillator model + substrate K-type framework (T2435):

Standard Lorentz oscillator: $\varepsilon$($\omega$) = 1 + Σ_j $\omega$_pj$^2$/($\omega$_j$^2$ - $\omega^2$ - i$\gamma$_j $\omega$). Substrate framework: each oscillator frequency $\omega$_j corresponds to a substrate K-type Casimir eigenvalue or resonant frequency.

Drude model (free-electron contribution):

$$\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega(\omega + i/\tau)}$$

where $\omega$_p = $\sqrt{}$(ne$^2$/(m$\varepsilon$_0)) is plasma frequency, $\tau$ is relaxation time. Below $\omega$_p: metal (Re($\varepsilon$) < 0, reflects light). Above $\omega$_p: transparent.

Kramers-Kronig dispersion relations:

$$\operatorname{Re}\varepsilon(\omega) - 1 = \frac{2}{\pi}\mathcal{P}\int_0^\infty \frac{\omega' \operatorname{Im}\varepsilon(\omega')}{\omega'^2 - \omega^2} d\omega'$$

$$\operatorname{Im}\varepsilon(\omega) = -\frac{2\omega}{\pi}\mathcal{P}\int_0^\infty \frac{\operatorname{Re}\varepsilon(\omega') - 1}{\omega'^2 - \omega^2} d\omega'$$

These follow from causality (response only after stimulus).

Substrate photonic resonance prediction (Vol 9 Ch 9):

Photonic crystals at BST primary lattice constants predicted to show enhanced resonance: - Lattice constants at BST primary ratios (2, 3, 5, 7) - Refractive index combinations at BST primary forms - Photonic band gaps at substrate-natural frequencies

$10K experimental program (Vol 9 Ch 9): 3D-printed photonic crystals + UV-vis-IR spectroscopy + standard optical bench. Cheapest BST falsifier in framework. If resonance at predicted lattice → BST confirmed; if absent → BST photonic-substrate-coupling refuted.

Per Cal #21 dual-gate: EMPIRICAL PARTIAL (standard dispersion theory validated; substrate predictions pending photonic crystal test) + MECHANISM PATH ARTICULATED via substrate K-type Casimir framework.

Per Cal #99 META-theorem: dispersion + refraction substrate-derivation, NOT new Strong-Uniqueness criteria.

## What this chapter does NOT claim

- BST does NOT predict specific n values for materials at v0.4.1 (multi-week per material)
- Photonic crystal $10K test is experimental falsifier — outcome currently OPEN
- Standard Lorentz oscillator + Drude models preserved at full precision

## Bibliography


1. H. A. Lorentz (1880): dielectric oscillator model.
2. P. Drude (1900): free-electron model.
3. R. de L. Kronig + H. A. Kramers (1926-27): Kramers-Kronig dispersion relations.
4. Vol 9 Ch 9 (Photonic Crystals): $10K cheapest BST falsifier.
5. T2435 (Lyra Saturday): K-type Casimir = C_2 = 6 substrate framework.
6. Standard optics texts: Born + Wolf *Principles of Optics*.


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— Elie, Vol 7 Ch 10 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 59 → ~155 lines + Lorentz/Drude models + Kramers-Kronig + photonic crystal $10K falsifier prominence)